SHAGUN AGARWAL

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SCHOLASTIC RECORD

Program	Institute	%/CGPA	Year
Bachelor and Master of Technology in Naval Architecture Ocean Engineering & Applied Mechanics; Minor in Systems Engineering	Indian Institute of Technology Madras, Chennai	8.71/10	2017
All India Senior School Exam	CMR NPS, Bangalore	94.6/100	2012
All India Secondary School Exam	CMR NPS, Bangalore	9.8/10	2010

RESEARCH INTERESTS

Computational Fluid Dynamics

- Finite element method
- Mesh-free methods
- Hybrid modelling

Coastal and Ocean Engineering

- Near shore surface waves
- Fluid-structure interaction
- Water impact problems
- Turbulence in breaking waves
- Ship-generated waves
- Storm surge due to cyclones

SKILLS

- Simulation: ANSYS Fluent, StarCCM+, OpenFOAM, ICEM-CFD
- Analysis & Presentation: MATLAB, Mathematica, Tecplot, Paraview
- Documentation: LaTex, MS Office, Adobe Photoshop, Markdown
- Programming Languages: C, C++, FORTRAN, Python; proficient at Linux and Windows OS
- 3D Modelling: AutoCAD, SolidWorks, Blender
- Experience with AVR family micro-controllers and ARM architecture microprocessor Raspberry-Pi
- Experience in working with large teams and managing funds for long duration projects

PUBLICATIONS

[1]Agarwal S, Sriram V, Yan S, Murali K; (2021); "Improvements in MLPG formulation for 3D wave interaction with fixed structures" in Computers and Fluids, 218; Elsevier

[2]Agarwal S, Sriram V, Murali K; (2019); "Interaction of Fixed Cylinder With Waves Through Weakly Coupled FNPT and Lagrangian Navier-Stokes" in Proceedings of the ASME 2019 38th International Conference on Ocean, Offshore and Arctic Engineering. Volume 7A: Ocean Engineering. UK.

[3]Agarwal S, Sriram V, Murali K; (2019); "Modelling wave interaction with porous structures using Boussinesq Equation" in Proceedings of International Conference in Ocean Engineering 2018; Springer

[4]Agarwal S, Jhunjhunwala T, Saikiran NT, Rajagopal P; (2018); "Demonstration of control and inspection by circumferentially guided ultrasonic waves using novel Remotely Operated Underwater Vehicle" in ISSS Journal of Micro and Smart Systems; Springer

SCHOLASTIC ACHIEVEMENTS

- Branch position 1 in Applied Mechanics dual degree program (2017) out of 15 students
- Secured All-India rank of 3322 (Top 0.6%) out of nearly 500,000 students in IIT-JEE 2012
- Secured All-India rank of 872 (Top 0.1%) out of nearly one million students in AIEEE 2012
- One of 220 awarded the Kishore Vaigyanik Protsahan Yojana research fellowship in 2011 by DST
- Awarded state level National Talent Search Examination scholarship in 2008 by NCERT
- One among 135 research scholars awarded the Prime Minister Research Fellowship in 2018.

CONFERENCES AND VISITS

- Coordinated in the session organised by Dr. V. Sriram on "Comparative Study on Interaction of Focused Waves with Fixed and Moving Cylinder" for international conference ISOPE-2019, Shanghai

- Selected, organised and distributed the cases for comparative study among 20+ participants from universities in UK, China, India, New Zealand and South Korea
- Systematically collected, compiled and visualised the 20+ submitted results for the comparative study. Co-authored the final comparative study journal publications.

- Visited City, University of London, UK to work under the guidance of Dr. Shiqiang Yan during the period Jun-Sep 2019. Worked on developing 3D MLPGR and learning their 3D qaleFOAM model.
- Visited University College London, UK to work with Prof. Thorsten Stoesser during the period Jun-Sep 2019. Learned Hydro3D, the in-house LES code for two phase with structure implemented using IBM. Simulated interaction of steep focussing waves with cylinder to study turbulent wake structures.
- Participated in 1st CCP-WSI Hackathon during 16-20 September, 2019 at Portaferry, Northern Ireland. Interacted with leading researchers part of UK Fluids Network's Special Interest Group in Wave Structure Interaction. Was added as member of CCP-WSI code repository, learned to use and develop the OpenFOAM based models. Exposed to current state of research and future domains of focus.

RESEARCH EXPERIENCE

Boussinesq Model for Modelling Water-Waves [↗] (Jun 2017 – Present)

Part of PhD Thesis Project guided by **Dr. V Sriram** [↗] **and Prof. K Murali** [↗]

Aim	- Develop a FEM model to use enhanced Boussinesq equations to simulate surface waves in nearshore region, ports and harbours, and inland waterways.
Method	- Evaluated complete finite element formulation using standard Galerkin method with mixed spatial interpolation using linear and quadratic triangular elements and analytical integration. - FORTRAN code (OpenMP) for unstructured mesh capable of handling irregular geometries.
Result	- Successfully verified the model for non-linear refraction-diffraction of regular waves using benchmark Berkhoff shoal test, and for flow through porous breakwater using experiments [3] - The model can also simulate waves generated by multiple vessels moving along arbitrary paths. It can estimate wave-making force on each vessel along surge and sway. - The water surface model can evaluate depth resolved velocities using mesh-free methods. Useful for coupling with 3D Navier-Stokes models and estimating sediment transport.

Wave-Structure Interaction using MLPGR [↗] (Jun 2017 – Present)

Part of PhD Thesis Project guided by **Dr. V Sriram** [↗] **and Prof. K Murali** [↗]

Aim	- Develop a 3D particle based mode for Navier-Stokes equations using meshless local Petrov-Galerkin method (MLPG) for wave interaction with fixed and floating structures.
Method	- Built a 3D formulation upon existing MLPG with Rankine source (MLPGR). The pressure Poisson equation solved in local weak-form with semi-analytical integration. - Wrote a modular OpenMP parallel FORTRAN code for the formulation of 3D MLPGR. - New free-surface detection, optimised neighbour search, water-line treatment, 3D structure boundary implementation, parallel solver and Paraview output.
Result	- Conducted numerical study on various stability criterion, particle resolution for 3D MLPGR. - Coupled with 2D-FNPT model for wave-generation, thus saving computational effort. - Verified the model using interaction of steep focusing waves and steep regular waves with fixed cylinder against experimental pressure-probe and wave-probe results. Publication [1,2] - Hybrid model coupling MLPGR with Boussinesq-FEM for efficient large domain simulation.

Fast SWE-FEM Model for Storm Surge [↗] (Apr 2019 – Present)

Research Project guided by **Dr. V Sriram** [↗]

Aim	- Develop FEM model for shallow water equation to get storm surge from tropical cyclones
Method	- FEM model based on lumped mass matrix, hence avoids any matrix inversion operation. - Symmetric and asymmetric formulae for wind velocity. Cyclone path can be arbitrary. - GPU parallel implemented using OpenACC standard. 20x speedup on Tesla series GPU.
Result	- Verified SWE using Airy wave theory. Tested cyclone Gaja, Vardah in Bay of Bengal.

Immersed Boundary Method for Fluid Structure Interaction [↗] (Aug 2016 – Jul 2017)

Graduate Thesis Project guided by **Prof. K Murali** [↗] **and Dr. K Arul Prakash** [↗]

Aim	- Implement Immersed Boundary Method (IBM) in a finite-element fluid solver.

Method	- Followed Immersed Finite Element Method [5] for implementation on a fully unstructured mesh using CBS [6] based fluid physics model, Moving Least Square interpolation function (MLS) and plane stress/strain based solid physics model.
Result	- Successfully verified CBS and MLS implementation through standard test cases.

Underwater Robotic Systems for Pipeline and Tank Inspections [↗] (Aug 2014 – Mar 2016)
Undergraduate Research Project guided by **Dr. Prabhu Rajagopal** [↗]

Aim	- To develop underwater robotic systems for inspecting submerged pipeline networks and metal floors of large industrial tanks for faults and cracks using methods of non-destructive testing.
Method	- Developed electronics and software support for sensors and actuator, implemented control and navigation systems, using cost effective microprocessor Raspberry-Pi. - Made a general-purpose Linux support from scratch for the ultrasonic transducer.
Result	- A remotely operated underwater vessel for inspecting pipeline network. Publication [4] - A submersible crawler for inspecting industrial tank floors.

Professional Experience

Planys Technologies [↗] India's first underwater ROV company Chennai, India
- Created a program for generating 3D profiles of screw propellers for ROVs, based on Kaplan series.
- A Solidworks compatible output is created for a blade based on diameter, number of blades, pitch by diameter ratio, and projected to expanded area ratio.
- Assisted in design and testing of thruster based on this propeller. Obtained up-to 80N of thrust from a 100mm propeller using 500W motor, which is competitive with the limited market options.

Internship at Cochin Shipyard Limited Kochi, India
India's Largest Shipyard with world class infrastructure (Dec 2014 – Jan 2015)
- Gained theoretical knowledge and on-site experience in shipbuilding, repair, design and planning and learned about standard industrial practices, equipment and routine at shipyard.
- Visited & studied navy aircraft carrier INS Vikrant, navy replenishment ship INS Aditya, amphibious warfare ship INS Shardul, passenger vessel MV Arabian Sea and coast guard's FPV-510 vessel

Internship at Forbes Marshall Chennai, India
One of India's Largest Energy Audit Companies (May 2014 – Jul 2014)
- Developed an energy model for manufacturing process of Autoclaved Aerated Cement and used it for energy audit at HIL Limited, Chennai, resulting in 5% increase in efficiency
- Prepared a mathematical model for on-site setup of steel curing chamber for Lanco Steel, Chennai. Identified 18 design parameters and their optimised set-points for higher output in new models.

Positions of Responsibility

Events Head – Workshops, Shaastra 2015 [7] (May 2014 – Jan 2015)
- Part of core team which organised 60+ events with 180+ coordinators and 25,000+ footfall
- Personally selected, guided 50 member team over 10 months for conducting 28 workshop in fields of robotics, systems engineering, automobile engineering, website & mobile app development, etc.

Coordinator – Robotics Workshops, Shaastra 2014 [7] (May 2013 – Jan 2014)
- Organised & conducted hands-on autonomous and manual robotics workshop for over 800 participants
- Personally handled budget of $8000 for over 300 custom kits, lecture material and venue

Social and Co-curricular Activities

National Service Scheme Volunteer (Aug 2013 – Jun 2014)
- Member of Computer Literacy For All project for the under-privileged school students
- Taught basics of using a computer and internet through regular hand-on sessions

ABU Robocon 2013, 2014 (Sep 2012 – Mar 2014)
- Part of 25-member team representing IIT Madras at the national level robotics event for two years.
- Worked in electronics and control division on development of the host 'three-transwheel omni-directional self-navigating robot' with positioning based on dead-reckoning

[5] Zhang, Gerstenberge et al; IFEM [6] Zienkiewicz, Nithiarasu et al; CBS [7] Annual technical festival of IITM

For more details, please visit this link.